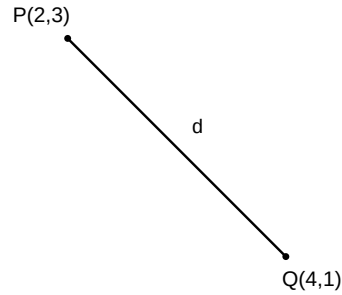


Diag Solutions

1. 1. Find the distance between the points $P(2, 3)$ and $Q(4, 1)$.



Answer:

$$2\sqrt{2}\text{units}$$

Solution:

1. Identify coordinates $x_1 = 2, y_1 = 3$ and $x_2 = 4, y_2 = 1$.
2. Apply distance formula $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

Derivation:

$$\begin{aligned}d &= \sqrt{(4 - 2)^2 + (1 - 3)^2} \\d &= \sqrt{2^2 + (-2)^2} \\d &= \sqrt{4 + 4} \\d &= \sqrt{8} = 2\sqrt{2}\end{aligned}$$

Coordinate Geometry — Distance Formula

2. 2. Determine the ratio in which the point $P(x, 2)$ divides the line segment joining the points $A(12, 5)$ and $B(4, -3)$.

Answer:

$$3 : 5$$

Solution:

1. Use section formula for y-coordinate: $y = \frac{my_2 + ny_1}{m + n}$.
2. Substitute values and solve for the ratio $m : n$.

Derivation:

$$\begin{aligned}
2 &= \frac{m(-3) + n(5)}{m + n} \\
2m + 2n &= -3m + 5n \\
5m &= 3n \\
\frac{m}{n} &= \frac{3}{5}
\end{aligned}$$

Coordinate Geometry — Section Formula

3. **3.** Find the coordinates of the midpoint of the line segment joining the points $A(-5, 4)$ and $B(7, -8)$.

Answer:

$$(1, -2)$$

Solution:

1. Apply midpoint formula $M = (\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2})$.
2. Calculate coordinates by substituting the given values.

Derivation:

$$\begin{aligned}
M &= (\frac{-5+7}{2}, \frac{4-8}{2}) \\
M &= (\frac{2}{2}, \frac{-4}{2}) \\
M &= (1, -2)
\end{aligned}$$

Coordinate Geometry — Midpoint Formula

4. **4.** Find the value of k if the point $P(0, 2)$ is equidistant from $A(3, k)$ and $B(k, 5)$.

Answer:

$$k = 1$$

Solution:

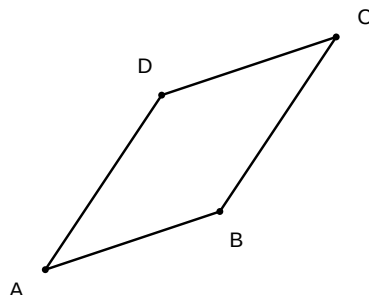
1. Use the condition $PA^2 = PB^2$ since $PA = PB$.
2. Expand and solve the resulting equation for k .

Derivation:

$$\begin{aligned}
(3-0)^2 + (k-2)^2 &= (k-0)^2 + (5-2)^2 \\
9 + k^2 - 4k + 4 &= k^2 + 9 \\
-4k + 4 &= 0 \\
k &= 1
\end{aligned}$$

Coordinate Geometry — Distance Formula Application

5. 5. If the coordinates of the vertices of a parallelogram ABCD are $A(1, 2)$, $B(4, 3)$, $C(6, 6)$, find the coordinates of the fourth vertex $D(x, y)$.



Answer:

$$(3, 5)$$

Solution:

1. Use the property that diagonals of a parallelogram bisect each other.
2. Equate midpoints of AC and BD .

Derivation:

$$\text{Midpoint of } AC = \left(\frac{1+6}{2}, \frac{2+6}{2} \right) = (3.5, 4)$$

$$\frac{4+x}{2} = 3.5$$

$$\Rightarrow x = 3$$

$$\frac{3+y}{2} = 4$$

$$\Rightarrow y = 5$$

Coordinate Geometry — Properties of Quadrilaterals

6. 6. Find the ratio in which the y -axis divides the line segment joining the points $A(5, -6)$ and $B(-1, -4)$. Also, find the point of intersection.

Answer:

$$\text{Ratio } 5 : 1, \text{ Point } (0, -13/3)$$

Solution:

1. Assume the ratio is $k : 1$. The coordinates of the point on the y -axis will be $(0, y)$.
2. Use the section formula for the x -coordinate to find k .
3. Substitute k into the section formula for the y -coordinate to find the point.

Derivation:

$$0 = \frac{k(-1) + 1(5)}{k + 1}$$

$$\Rightarrow -k + 5 = 0$$

$$\Rightarrow k = 5$$

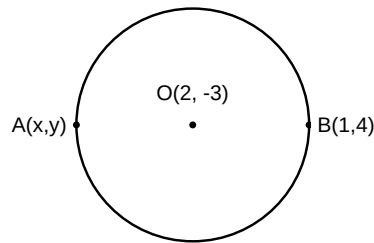
Ratio is 5 : 1

$$y = \frac{5(-4) + 1(-6)}{5 + 1} = \frac{-20 - 6}{6} = \frac{-26}{6} = -\frac{13}{3}$$

Point is $(0, -\frac{13}{3})$

Coordinate Geometry — Section Formula

7. 7. Find the coordinates of a point A , where AB is the diameter of a circle whose centre is $(2, -3)$ and B is $(1, 4)$.



Answer:

$$A(3, -10)$$

Solution:

1. Let $A = (x, y)$. The centre $(2, -3)$ is the midpoint of the diameter AB .
2. Apply the midpoint formula: $x_m = \frac{x_1 + x_2}{2}$ and $y_m = \frac{y_1 + y_2}{2}$.
3. Solve for x and y .

Derivation:

$$\frac{x + 1}{2} = 2$$

$$\Rightarrow x + 1 = 4$$

$$\Rightarrow x = 3$$

$$\frac{y + 4}{2} = -3$$

$$\Rightarrow y + 4 = -6$$

$$\Rightarrow y = -10$$

$$A = (3, -10)$$

Coordinate Geometry — Midpoint Formula

8. **8.** Find the value of k for which the points $A(2, 3)$, $B(4, k)$, and $C(6, -3)$ are collinear.

Answer:

$$k = 0$$

Solution:

1. For three points to be collinear, the area of the triangle formed by them must be 0.
2. Use the area formula: $\frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)| = 0$.
3. Substitute the coordinates and solve for k .

Derivation:

$$\begin{aligned}2(k - (-3)) + 4(-3 - 3) + 6(3 - k) &= 0 \\2(k + 3) + 4(-6) + 18 - 6k &= 0 \\2k + 6 - 24 + 18 - 6k &= 0 \\-4k &= 0 \\\Rightarrow k &= 0\end{aligned}$$

Coordinate Geometry — Area of Triangle

9. **9.** Find the distance between the points $P(-6, 7)$ and $Q(-1, -5)$.

Answer:

$$13\text{units}$$

Solution:

1. Use the distance formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.
2. Substitute the coordinates of P and Q .
3. Calculate the square roots to find the distance.

Derivation:

$$\begin{aligned}d &= \sqrt{(-1 - (-6))^2 + (-5 - 7)^2} \\d &= \sqrt{(5)^2 + (-12)^2} \\d &= \sqrt{25 + 144} = \sqrt{169} \\d &= 13 \text{ units}\end{aligned}$$

Coordinate Geometry — Distance Formula

10. **10.** If the points $A(6, 1)$, $B(8, 2)$, $C(9, 4)$, and $D(p, 3)$ are the vertices of a parallelogram taken in order, find the value of p .

Answer:

$$p = 7$$

Solution:

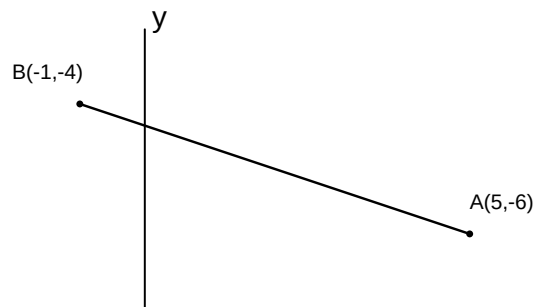
1. Diagonals of a parallelogram bisect each other, meaning their midpoints coincide.
2. Find the midpoint of diagonal AC and diagonal BD .
3. Equate the midpoints and solve for p .

Derivation:

$$\begin{aligned}\text{Midpoint of } AC &= \left(\frac{6+9}{2}, \frac{1+4}{2}\right) = (7.5, 2.5) \\ \text{Midpoint of } BD &= \left(\frac{8+p}{2}, \frac{2+3}{2}\right) = \left(\frac{8+p}{2}, 2.5\right) \\ \frac{8+p}{2} &= 7.5 \\ \Rightarrow 8+p &= 15 \\ \Rightarrow p &= 7\end{aligned}$$

Coordinate Geometry — Midpoint Formula

11. 11. Find the ratio in which the y -axis divides the line segment joining the points $A(5, -6)$ and $B(-1, -4)$. Also, find the point of intersection.

**Answer:**

Ratio is 5 : 1 and the point of intersection is $(0, -13/3)$.

Solution:

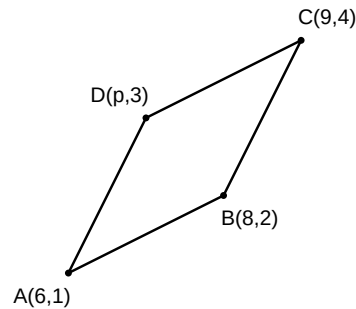
1. Let the ratio be $k : 1$.
2. The coordinates of the point on the y -axis are $(0, y)$.
3. Use the section formula for the x -coordinate to find k .
4. Substitute k into the section formula to find the y -coordinate.
5. State the final ratio and the point.

Derivation:

$$\begin{aligned}
 x &= \frac{m_1x_2 + m_2x_1}{m_1 + m_2} \\
 \Rightarrow 0 &= \frac{k(-1) + 1(5)}{k + 1} \\
 \Rightarrow -k + 5 &= 0 \\
 \Rightarrow k &= 5 \\
 \text{Ratio is } 5 : 1 \\
 y &= \frac{m_1y_2 + m_2y_1}{m_1 + m_2} = \frac{5(-4) + 1(-6)}{5 + 1} = \frac{-20 - 6}{6} = \frac{-26}{6} = -\frac{13}{3} \\
 \text{Point is } &(0, -\frac{13}{3})
 \end{aligned}$$

Coordinate Geometry — Section Formula

- 12. 12.** If the points $A(6, 1)$, $B(8, 2)$, $C(9, 4)$, and $D(p, 3)$ are the vertices of a parallelogram taken in order, find the value of p .



Answer:

$$p = 7$$

Solution:

1. Use the property that diagonals of a parallelogram bisect each other.
2. Find the midpoint of diagonal AC .
3. Find the midpoint of diagonal BD .
4. Equate the midpoints since they represent the same point.
5. Solve for p .

Derivation:

$$\begin{aligned}
 \text{Midpoint of } AC &= \left(\frac{6+9}{2}, \frac{1+4}{2} \right) = (7.5, 2.5) \\
 \text{Midpoint of } BD &= \left(\frac{8+p}{2}, \frac{2+3}{2} \right) = \left(\frac{8+p}{2}, 2.5 \right) \\
 \frac{8+p}{2} &= 7.5 \\
 \Rightarrow 8+p &= 15 \\
 \Rightarrow p &= 7
 \end{aligned}$$

- 13. 13.** Find the area of a triangle whose vertices are $(1, -1)$, $(-4, 6)$, and $(-3, -5)$.

Answer:

24square units.

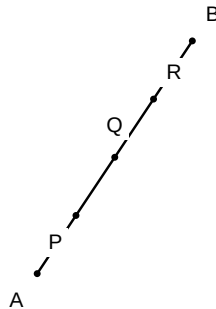
Solution:

1. Identify the coordinates $(x_1, y_1), (x_2, y_2), (x_3, y_3)$.
2. Use the area of triangle formula: $\frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$.
3. Substitute the values into the formula.
4. Perform the arithmetic calculation.
5. Express the final result in square units.

Derivation:

$$\begin{aligned}
 \text{Area} &= \frac{1}{2}|1(6 - (-5)) + (-4)(-5 - (-1)) + (-3)(-1 - 6)| \\
 &= \frac{1}{2}|1(11) + (-4)(-4) + (-3)(-7)| \\
 &= \frac{1}{2}|11 + 16 + 21| \\
 &= \frac{1}{2}|48| = 24 \text{ Sq. units}
 \end{aligned}$$

- 14. 14.** Find the coordinates of the points which divide the line segment joining $A(-2, 2)$ and $B(2, 8)$ into four equal parts.



Answer:

$(-1, 3.5), (0, 5), \text{ and } (1, 6.5)$.

Solution:

1. Identify that four equal parts require three points of division P, Q, R .

- Point Q is the midpoint of AB .
- Point P is the midpoint of AQ .
- Point R is the midpoint of QB .
- Calculate each midpoint using the midpoint formula.

Derivation:

$$Q = \left(\frac{-2+2}{2}, \frac{2+8}{2} \right) = (0, 5)$$

$$P = \left(\frac{-2+0}{2}, \frac{2+5}{2} \right) = (-1, 3.5)$$

$$R = \left(\frac{0+2}{2}, \frac{5+8}{2} \right) = (1, 6.5)$$

Coordinate Geometry — Section Formula

- 15. 15.** If the circumference of a circle is 132 cm, then its radius is:

Answer: (B)

Solution:

- Recall the formula for circumference $C = 2\pi r$.
- Substitute the given value $132 = 2 \times \frac{22}{7} \times r$.
- Solve for r .

Derivation:

$$C = 2\pi r$$

$$\Rightarrow 132 = 2 \times \frac{22}{7} \times r$$

$$\Rightarrow 132 = \frac{44}{7}r$$

$$\Rightarrow r = \frac{132 \times 7}{44}$$

$$\Rightarrow r = 3 \times 7 = 21 \text{ cm}$$

Areas Related to Circles — Circumference of a circle

- 16. 16.** The area of a circle whose diameter is 14 cm is:

Answer: (C)

Solution:

- Find the radius $r = \frac{d}{2} = \frac{14}{2} = 7 \text{ cm}$.
- Use the area formula $A = \pi r^2$.
- Calculate $A = \frac{22}{7} \times 7^2$.

Derivation:

$$r = 7 \text{ cm}$$

$$\Rightarrow A = \frac{22}{7} \times 7 \times 7 = 22 \times 7 = 154 \text{ cm}^2$$

17. 17. The area of a quadrant of a circle with radius $r = 7$ cm is:

Answer: (A)

Solution:

1. A quadrant is $\frac{1}{4}$ of a circle.
2. Calculate total area $A = \pi r^2 = \frac{22}{7} \times 7^2 = 154 \text{ cm}^2$.
3. Divide by 4.

Derivation:

$$\text{Area} = \frac{1}{4}\pi r^2 = \frac{1}{4} \times \frac{22}{7} \times 7 \times 7 = \frac{154}{4} = 38.5 \text{ cm}^2$$

18. 18. If the perimeter of a semi-circular protractor is 36 cm, find its diameter.

Answer: (B)

Solution:

1. The perimeter of a semicircle is $\pi r + 2r$.
2. Set $\pi r + 2r = 36$.
3. Solve for r and then find $d = 2r$.

Derivation:

$$\begin{aligned} r(\pi + 2) &= 36 \\ \Rightarrow r\left(\frac{22}{7} + 2\right) &= 36 \\ \Rightarrow r\left(\frac{36}{7}\right) &= 36 \\ \Rightarrow r &= 7 \\ \Rightarrow d = 2r &= 14 \text{ cm} \end{aligned}$$

19. 19. The area of a sector of a circle with radius 6 cm and central angle 60° is:

Answer: (D)

Solution:

1. Use the formula $\text{Area} = \frac{\theta}{360} \times \pi r^2$.
2. Substitute $\theta = 60$ and $r = 6$.
3. Calculate the result.

Derivation:

$$\text{Area} = \frac{60}{360} \times \pi \times 6^2 = \frac{1}{6} \times \pi \times 36 = 6\pi \text{ cm}^2$$

- 20. 20.** Assertion (A): The area of a circle with radius 7 cm is 154 cm^2 . Reason (R): The area of a circle is given by the formula πr^2 , where r is the radius.

Answer: (A)

Solution:

1. Recall formula πr^2 .
2. Substitute $r = 7$ and $\pi = 22/7$.
3. Calculate $22/7 \times 7 \times 7 = 154$.

Derivation: Area = πr^2

$$\text{Area} = \frac{22}{7} \times 7 \times 7$$

$$\text{Area} = 22 \times 7 = 154 \text{ cm}^2$$

Areas Related to Circles — Area of a Circle

- 21. 21.** Assertion (A): The circumference of a circle with diameter 14 cm is 44 cm. Reason (R): The circumference of a circle is given by πd , where d is the diameter.

Answer: (A)

Solution:

1. Recall formula $C = \pi d$.
2. Substitute $\pi = 22/7$ and $d = 14$.
3. Calculate $22/7 \times 14 = 44$.

Derivation:

$$C = \pi d$$

$$C = \frac{22}{7} \times 14$$

$$C = 22 \times 2 = 44 \text{ cm}$$

Areas Related to Circles — Circumference of a Circle

- 22. 22.** Assertion (A): The area of a semicircle with radius r is $\frac{1}{2}\pi r^2$. Reason (R): A semicircle is half of a full circle.

Answer: (A)

Solution:

1. Area of full circle is πr^2 .
2. Semicircle is exactly half of a circle.
3. Divide the area formula by 2.

Derivation:

$$\text{Area of semicircle} = \frac{1}{2} \times \text{Area of circle}$$

$$\text{Area of semicircle} = \frac{1}{2}\pi r^2$$

- 23. 23.** Assertion (A): The length of an arc of a circle with radius r and central angle θ is $\frac{\theta}{360} \times 2\pi r$. Reason (R): The circumference of a circle is $2\pi r$.

Answer: (A)

Solution:

1. Arc length is a fraction of the circumference.
2. The fraction is determined by the central angle θ over 360° .
3. Combine to get $\frac{\theta}{360} \times 2\pi r$.

Derivation:

$$\text{Arc Length} = \frac{\theta}{360^\circ} \times 2\pi r$$

- 24. 24.** Assertion (A): The area of a quadrant of a circle with radius r is $\frac{1}{4}\pi r^2$. Reason (R): A quadrant is a sector with a central angle of 45° .

Answer: (C)

Solution:

1. The area of a quadrant is indeed $\frac{1}{4}\pi r^2$.
2. A quadrant implies a central angle of 90° , not 45° .
3. Therefore, Reason is false.

Derivation:

$$\text{Area of sector} = \frac{\theta}{360^\circ} \times \pi r^2$$

For quadrant, $\theta = 90^\circ$

$$\text{Area} = \frac{90}{360} \times \pi r^2 = \frac{1}{4}\pi r^2$$

- 25. 25.** Find the area of a sector of a circle with radius 6 cm if the angle of the sector is 60° .

Answer:

$$\frac{132}{7}\text{ cm}^2$$

Solution:

1. Use the formula for the area of a sector $A = \frac{\theta}{360^\circ} \times \pi r^2$.
2. Substitute $\theta = 60^\circ$, $r = 6\text{ cm}$, and $\pi = \frac{22}{7}$ to calculate the area.

Derivation:

$$\begin{aligned}
 A &= \frac{60^\circ}{360^\circ} \times \frac{22}{7} \times 6^2 \\
 \Rightarrow A &= \frac{1}{6} \times \frac{22}{7} \times 36 \\
 \Rightarrow A &= 6 \times \frac{22}{7} = \frac{132}{7} \text{ cm}^2
 \end{aligned}$$

Areas Related to Circles — Area of a Sector

- 26. 26.** The length of the minute hand of a clock is 14 cm. Find the area swept by the minute hand in 5 minutes.

Answer:

$$\frac{154}{3} \text{ cm}^2$$

Solution:

1. Calculate the angle swept by the minute hand in 5 minutes: since it sweeps 360° in 60 minutes, in 5 minutes it sweeps $\frac{360}{60} \times 5 = 30^\circ$.
2. Apply the area formula for a sector with $r = 14 \text{ cm}$ and $\theta = 30^\circ$.

Derivation:

$$\begin{aligned}
 \theta &= \frac{360^\circ}{60} \times 5 = 30^\circ \\
 \Rightarrow \text{Area} &= \frac{30^\circ}{360^\circ} \times \frac{22}{7} \times 14^2 \\
 \Rightarrow \text{Area} &= \frac{1}{12} \times \frac{22}{7} \times 196 = \frac{1}{12} \times 22 \times 28 = \frac{616}{12} = \frac{154}{3} \text{ cm}^2
 \end{aligned}$$

Areas Related to Circles — Area of a Sector

- 27. 27.** If the perimeter and the area of a circle are numerically equal, then find the radius of the circle.

Answer:

$$2 \text{ units}$$

Solution:

1. Set the formula for circumference $2\pi r$ equal to the formula for area πr^2 .
2. Solve the equation $2\pi r = \pi r^2$ for r .

Derivation:

$$\begin{aligned}
 2\pi r &= \pi r^2 \\
 \Rightarrow 2 &= r \text{ (since } r \neq 0)
 \end{aligned}$$

Areas Related to Circles — Area and Perimeter of a Circle

28. 28. Find the area of a quadrant of a circle whose circumference is 44 *cm*.

Answer:

$$38.5\text{cm}^2$$

Solution:

1. Use the circumference $2\pi r = 44$ to find the radius r .
2. Calculate the area of the quadrant using $\frac{1}{4}\pi r^2$.

Derivation:

$$\begin{aligned}2 \times \frac{22}{7} \times r &= 44 \\ \Rightarrow r &= 7 \text{ cm} \\ \Rightarrow \text{Area} &= \frac{1}{4} \times \frac{22}{7} \times 7^2 \\ \Rightarrow \text{Area} &= \frac{1}{4} \times 22 \times 7 = \frac{154}{4} = 38.5 \text{ cm}^2\end{aligned}$$

Areas Related to Circles — Area of a Quadrant

29. 29. The radius of two circles are 8 *cm* and 6 *cm* respectively. Find the radius of the circle having area equal to the sum of the areas of the two circles.

Answer:

$$10\text{cm}$$

Solution:

1. Set the area of the new circle πR^2 equal to $\pi r_1^2 + \pi r_2^2$.
2. Solve $R^2 = 8^2 + 6^2$ to find R .

Derivation:

$$\begin{aligned}\pi R^2 &= \pi(8^2) + \pi(6^2) \\ \Rightarrow R^2 &= 64 + 36 = 100 \\ \Rightarrow R &= \sqrt{100} = 10 \text{ cm}\end{aligned}$$

Areas Related to Circles — Area of Combined Circles

30. 30. Find the area of a sector of a circle with radius 6 *cm* if the angle of the sector is 60° . Use $\pi = 3.14$.

Answer:

$$18.84\text{cm}^2$$

Solution:

1. Identify the radius $r = 6 \text{ cm}$ and central angle $\theta = 60^\circ$.

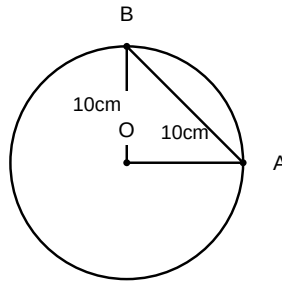
2. Use the formula for the area of a sector: $A = \frac{\theta}{360^\circ} \times \pi r^2$.
3. Substitute the values and calculate the final result.

Derivation:

$$\begin{aligned}
 A &= \frac{60}{360} \times 3.14 \times 6^2 \\
 A &= \frac{1}{6} \times 3.14 \times 36 \\
 A &= 3.14 \times 6 \\
 A &= 18.84 \text{ cm}^2
 \end{aligned}$$

Areas Related to Circles — Area of a Sector

- 31. 31.** In a circle of radius 10 cm, a chord AB subtends a right angle at the centre O . Find the area of the minor segment. (Use $\pi = 3.14$).



Answer:

$$28.5 \text{ cm}^2$$

Solution:

1. Calculate the area of the sector OAB with $\theta = 90^\circ$.
2. Calculate the area of $\triangle OAB$ which is a right-angled triangle with base and height equal to 10 cm.
3. Subtract the area of the triangle from the area of the sector to get the segment area.

Derivation:

$$\begin{aligned}
 \text{Area of sector} &= \frac{90}{360} \times 3.14 \times 10^2 = 78.5 \text{ cm}^2 \\
 \text{Area of } \triangle OAB &= \frac{1}{2} \times 10 \times 10 = 50 \text{ cm}^2 \\
 \text{Area of segment} &= 78.5 - 50 = 28.5 \text{ cm}^2
 \end{aligned}$$

Areas Related to Circles — Area of a Segment

- 32. 32.** The length of the minute hand of a clock is 14 cm. Find the area swept by the minute hand in 5 minutes.

Answer:

$$51.33\text{cm}^2$$

Solution:

1. Determine the angle swept in 60 minutes (360°), so in 5 minutes it is 30° .
2. Use the radius $r = 14$ cm.
3. Apply the sector area formula.

Derivation:

$$\begin{aligned}\theta &= \frac{360}{60} \times 5 = 30^\circ \\ \text{Area} &= \frac{30}{360} \times \frac{22}{7} \times 14 \times 14 \\ \text{Area} &= \frac{1}{12} \times 22 \times 2 \times 14 = \frac{616}{12} = 51.33 \text{ cm}^2\end{aligned}$$

Areas Related to Circles — Applications of Sector Area

- 33. 33.** The diameters of two circles are 8 cm and 6 cm respectively. Find the radius of a circle whose area is equal to the sum of the areas of the two circles.

Answer:

$$5\text{cm}$$

Solution:

1. Find the radii of the two circles: $r_1 = 4$ cm, $r_2 = 3$ cm.
2. Equate the sum of areas $\pi r_1^2 + \pi r_2^2$ to πR^2 .
3. Solve for R .

Derivation:

$$\begin{aligned}\pi(4)^2 + \pi(3)^2 &= \pi R^2 \\ 16\pi + 9\pi &= \pi R^2 \\ 25\pi &= \pi R^2 \\ R^2 &= 25 \\ \Rightarrow R &= 5 \text{ cm}\end{aligned}$$

Areas Related to Circles — Area of a Circle

- 34. 34.** A horse is tethered to a peg at one corner of a square-shaped grass field of side 15 m by means of a 5 m long rope. Find the area of that part of the field in which the horse can graze.

Answer:

$$19.625m^2$$

Solution:

1. Identify the shape as a quadrant of a circle with radius $r = 5 \text{ m}$.
2. The grazing area is the area of this quadrant.
3. Calculate $\frac{1}{4} \times \pi r^2$.

Derivation:

$$\begin{aligned}\text{Area} &= \frac{1}{4} \times 3.14 \times 5^2 \\ \text{Area} &= \frac{1}{4} \times 3.14 \times 25 \\ \text{Area} &= 0.785 \times 25 = 19.625 \text{ m}^2\end{aligned}$$

Areas Related to Circles — Area of a Sector

- 35. 35.** A round table cover of radius 28 cm has a design in the middle leaving an equilateral triangle ABC in the middle as shown in the figure. Find the area of the design (shaded region). Use $\sqrt{3} = 1.73$ and $\pi = \frac{22}{7}$.

Answer:

$$\text{Area of shaded region} = 1514.88 \text{ cm}^2$$

Solution:

1. Calculate the area of the circle with $r = 28 \text{ cm}$.
2. The triangle ABC is equilateral inscribed in the circle. The side length a of the equilateral triangle is $r\sqrt{3}$.
3. Calculate the area of the equilateral triangle using $\frac{\sqrt{3}}{4}a^2$.
4. Subtract the area of the triangle from the area of the circle to get the area of the shaded design.

Derivation: $\text{Area}_{\text{circle}} = \pi r^2 = \frac{22}{7} \times 28 \times 28 = 2464 \text{ cm}^2$ \\\ Side \; a = 28\sqrt{3} \text{ cm} \\\ \text{Area}_{\triangle} = \frac{\sqrt{3}}{4}(28\sqrt{3})^2 = \frac{1.73}{4} \times 784 \times 3 = 1.73 \times 196 \times 3 = 1017.36 \text{ cm}^2 \\\ \text{Area}_{\text{shaded}} = 2464 - 1017.36 = 1446.64 \text{ cm}^2

Areas Related to Circles — Area of Segment

- 36. 36.** A car has two wipers which do not overlap. Each wiper has a blade of length 25 cm sweeping through an angle of 115° . Find the total area cleaned at each sweep of the blades. (Take $\pi = \frac{22}{7}$)

Answer:

$$\text{Total area} = \frac{63250}{63} \text{ cm}^2 \text{ or approx } 1003.97 \text{ cm}^2$$

Solution:

1. Identify that each wiper sweeps a sector of a circle.

- Radius $r = 25 \text{ cm}$ and central angle $\theta = 115^\circ$.
- Area of one sector is $\frac{\theta}{360^\circ} \times \pi r^2$.
- Multiply the area of one sector by 2 since there are two wipers.

Derivation: Area = $2 \times \left(\frac{115}{360} \times \frac{22}{7} \times 25 \times 25 \right)$

$$\text{Area} = 2 \times \left(\frac{23}{72} \times \frac{22}{7} \times 625 \right)$$

$$\text{Area} = \frac{23 \times 11 \times 625}{36 \times 7}$$

$$\text{Area} = \frac{158125}{252} \approx 627.48 \times 2 = 1254.96 \text{ cm}^2$$

Areas Related to Circles — Area of Sector

- 37. 37.** In a circular park of radius 21 m , a path of uniform width 3.5 m is constructed inside the park along the boundary. Find the cost of paving the path at the rate of Rs.50 per m^2 .

Answer:

$$\text{Cost} = \text{Rs.}20625$$

Solution:

- The path forms an annulus (ring) between two circles.
- Outer radius $R = 21 \text{ m}$.
- Inner radius $r = R - 3.5 = 17.5 \text{ m}$.
- Area of path = $\pi(R^2 - r^2)$.
- Multiply the area by the rate 50.

Derivation: Area = $\frac{22}{7}(21^2 - 17.5^2)$

$$\text{Area} = \frac{22}{7}(21 - 17.5)(21 + 17.5)$$

$$\text{Area} = \frac{22}{7}(3.5)(38.5)$$

$$\text{Area} = 22 \times 0.5 \times 38.5 = 11 \times 38.5 = 423.5 \text{ m}^2$$

$$\text{Cost} = 423.5 \times 50 = 21175$$

Areas Related to Circles — Area of Annulus

- 38. 38.** A chord AB of a circle of radius 15 cm subtends an angle of 60° at the centre. Find the areas of the corresponding minor and major segments of the circle. (Use $\pi = 3.14$ and $\sqrt{3} = 1.73$)

Answer:

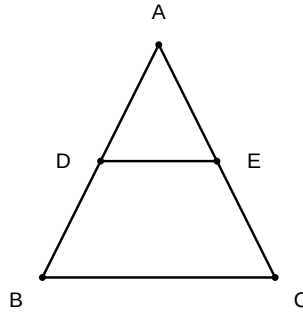
$$\text{Minor segment} = 20.4375 \text{ cm}^2, \text{ Major segment} = 686.0625 \text{ cm}^2$$

Solution:

- Calculate the area of the minor sector with $\theta = 60^\circ$.
- Calculate the area of $\triangle OAB$ where $OA = OB = 15$ and $\angle O = 60^\circ$ (which is equilateral).
- Minor segment area = Area of sector - Area of $\triangle OAB$.
- Major segment area = Area of circle - Area of minor segment.

Derivation: $\text{Area}_{\{\text{sector}\}} = \frac{60}{360} \times 3.14 \times 15^2 = \frac{1}{6} \times 3.14 \times 225 = 117.75 \text{ cm}^2$ \\
 $\text{Area}_{\{\triangle\}} = \frac{\sqrt{3}}{4} \times 15^2 = \frac{1.73}{4} \times 225 = 97.3125 \text{ cm}^2$ \\
 $\text{Area}_{\{\text{minor}\}} = 117.75 - 97.3125 = 20.4375 \text{ cm}^2$ \\
 $\text{Area}_{\{\text{circle}\}} = 3.14 \times 225 = 706.5 \text{ cm}^2$ \\
 $\text{Area}_{\{\text{major}\}} = 706.5 - 20.4375 = 686.0625 \text{ cm}^2$ *Areas Related to Circles — Area of Segment*

- 39. 39.** In $\triangle ABC$, if $DE \parallel BC$ such that $AD = 2 \text{ cm}$, $DB = 3 \text{ cm}$, and $AE = 4 \text{ cm}$, find the length of EC .



Answer: (C)

Solution:

1. Since $DE \parallel BC$, by Thales Theorem (BPT), we have $\frac{AD}{DB} = \frac{AE}{EC}$.
2. Substitute the given values: $\frac{2}{3} = \frac{4}{EC}$.
3. Solve for EC : $2 \times EC = 12$, so $EC = 6 \text{ cm}$.

Derivation:

$$\begin{aligned} \frac{AD}{DB} &= \frac{AE}{EC} \\ \Rightarrow \frac{2}{3} &= \frac{4}{EC} \\ \Rightarrow 2 \times EC &= 12 \\ \Rightarrow EC &= 6 \end{aligned}$$

Triangles — Basic Proportionality Theorem

- 40. 40.** If $\triangle ABC \sim \triangle PQR$ and $\frac{AB}{PQ} = \frac{1}{3}$, then the ratio $\frac{\text{ar}(\triangle ABC)}{\text{ar}(\triangle PQR)}$ is:

Answer: (A)

Solution:

1. According to the area of similar triangles theorem, the ratio of areas is equal to the square of the ratio of their corresponding sides.
2. The ratio of sides is $\frac{1}{3}$.
3. Square of the ratio is $(\frac{1}{3})^2 = \frac{1}{9}$.

Derivation:

$$\frac{ar(\triangle ABC)}{ar(\triangle PQR)} = \left(\frac{AB}{PQ}\right)^2 = \left(\frac{1}{3}\right)^2 = \frac{1}{9}$$

Triangles — Areas of Similar Triangles